

Ahmad Farhat Portfolio

Data Engineer · ML Practitioner · AI Systems Architect · Researcher

American · UAE Golden Visa

+971-50-316-3664

ahmad.farhat.research@gmail.com

Google Scholar ↗

Portfolio ↗

Github code sample ↗

LANGUAGES

Python SQL JavaScript Bash R MATLAB C++ Java

ML & AI

PyTorch TensorFlow Scikit-learn Deep Learning Transformers VICReg NumPy / SciPy LangGraph RAG Vector Databases MCP Servers Ollama / Local LLMs Claude Code Claude Skills & Plugins

DATA ENGINEERING

PostgreSQL pgvector Oracle Pandas ETL Pipelines D3.js Bloomberg BLPAPI

MLOPS & INFRA

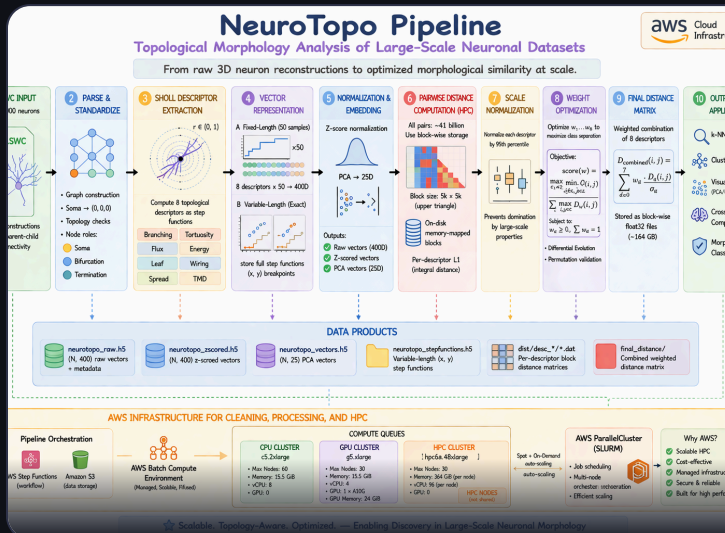
SLURM / HPC AWS Docker Git Linux FastAPI Apache Airflow HDF5 / h5py ChromaDB

FRAMEWORKS

Django Sentence-Transformers NetworkX eScriptorium PWA / Service Workers Chart.js

COMPUTATIONAL RESEARCH

Topological Data Analysis · Computational Neuroscience



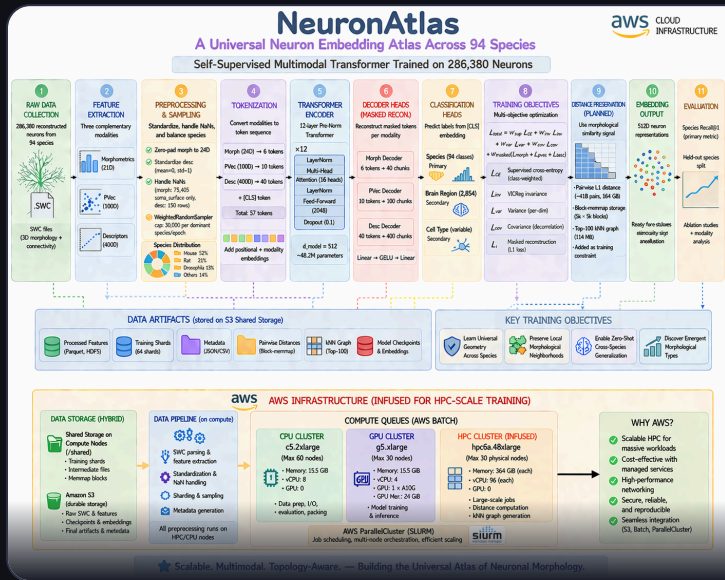
RESEARCH · TOPOLOGICAL ML · HPC

NeuroTopo — Large-Scale Neuronal Morphology Analysis

End-to-end HPC pipeline processing ~286,000 reconstructed neurons (SWC format) to compute 8 topological Sholl descriptors and produce a 286,000×286,000 pairwise distance matrix (~164 GB, ~41 billion unique pairs) with O(1) lookup. Descriptor weights optimized via differential evolution over labeled neuronal classes. Published in PLOS Computational Biology Q1 (25 citations).

Python NumPy / SciPy HDF5/h5py SLURM HPC A100 TDA NetworkX

► Technical architecture



RESEARCH · DEEP LEARNING · NATURE-TARGET

NeuronAtlas — Universal Neuron Embedding Atlas

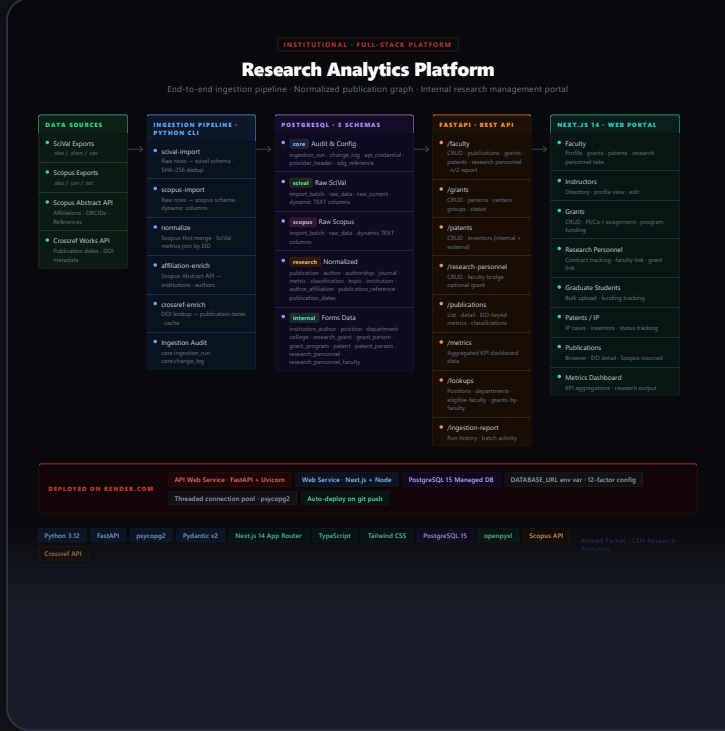
Self-supervised multimodal Transformer (12 layers, d=512, ~48.2M parameters) trained on 286,380 morphologically reconstructed neurons across 94 species. Learns a species-agnostic representation space enabling cross-species retrieval, morphological type discovery, and cell-type classification. Targeting Nature / Nature Methods publication.

PyTorch Transformers VICReg Multimodal AWS 286K neurons · 94 species

► Model & training details

INSTITUTIONAL ANALYTICS

Institutional Research & Analytics



INSTITUTIONAL · FULL-STACK · DATA ENGINEERING

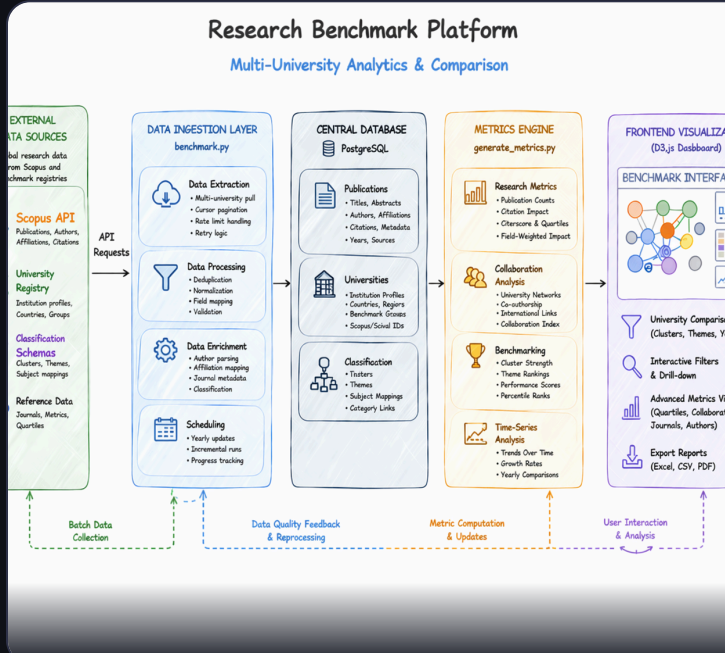
Institutional Research Analytics Platform

University-wide research intelligence platform integrating Scopus, SciVal, Crossref, and OpenAlex into a centralized PostgreSQL data warehouse across five purpose-built schemas. A Python ingestion pipeline normalizes raw vendor exports into structured entities with full diff-based idempotency and column-level provenance tracking. FastAPI backend exposes eight domain routers powering a Next.js 14 management portal with complete CRUD for faculty, research personnel, grants, and patents — deployed on Render.com.

PostgreSQL FastAPI Next.js 14 Scopus API Python ETL Render.com SciVal Crossref

► Architecture overview

[Live Demo](#) Login: demo / demo123
Hosted on Render. Initial load may take about a minute while the app server and database wake up.



INSTITUTIONAL · RESEARCH INTELLIGENCE

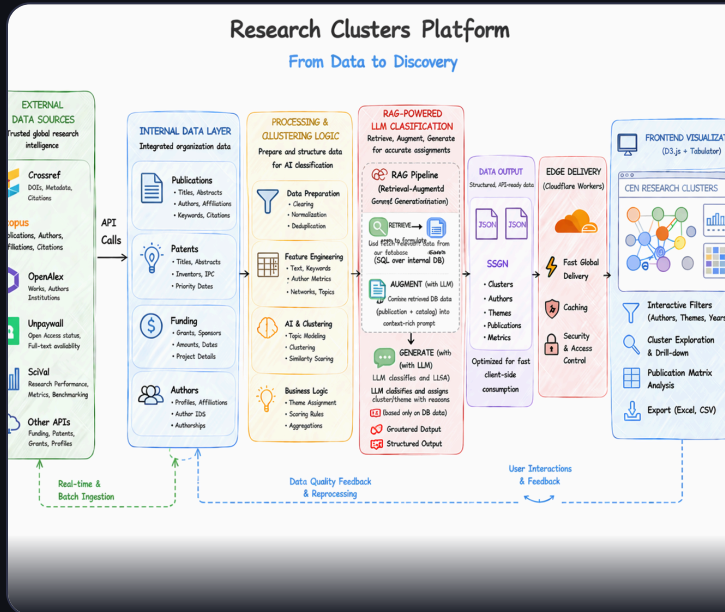
Research Analytics Benchmark Dashboard

University-wide research intelligence platform integrating Scopus, SciVal, OpenAlex, Crossref, and Unpaywall into a centralized PostgreSQL data warehouse. Provides senior leadership with live benchmarking against regional and global peer institutions — measuring publication output, journal quality (quartiles), collaboration networks, author productivity, and research trend forecasting — via interactive D3.js dashboards.

D3.js PostgreSQL Python ETL Scopus API OpenAlex Bibliometrics KPI frameworks

► What this platform does

[Live Demo](#)



INSTITUTIONAL · ML PIPELINE

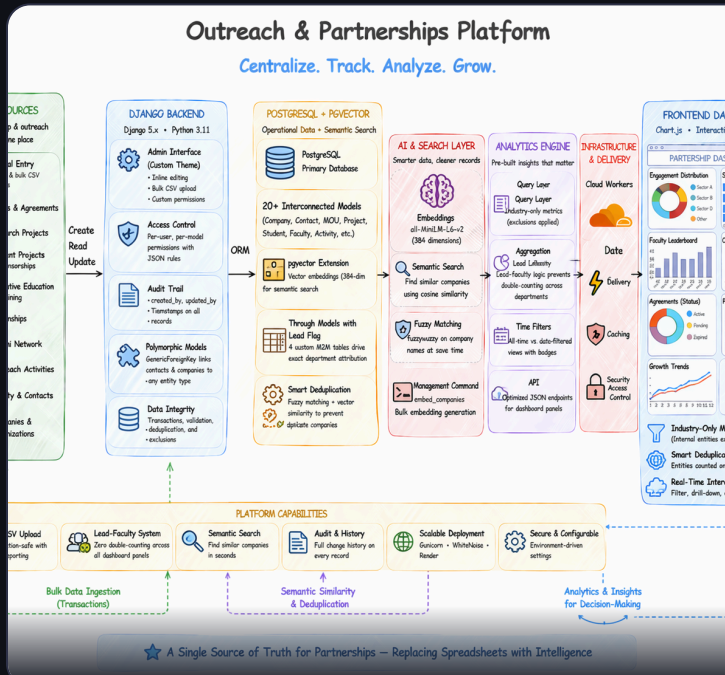
Research Clustering & Thematic Visualization

AI-powered pipeline applying dimensionality reduction (PCA) and unsupervised clustering to institutional publication corpora — mapping the research landscape to identify thematic strengths, gaps, and collaboration opportunities. Feeds into the benchmark dashboard for strategic foresight.

PCA Clustering Python pgvector NLP Visualization

► Pipeline overview

[Live Demo](#)



INSTITUTIONAL · FULL-STACK PLATFORM

Outreach Management Platform

Production full-stack Django/PostgreSQL application centralizing all academic outreach, industry partnerships, MOUs, research projects, senior design sponsorships, executive education, internships, and faculty/industry contacts across 6 engineering departments — replacing scattered spreadsheets with a single queryable platform with semantic search and interactive analytics.

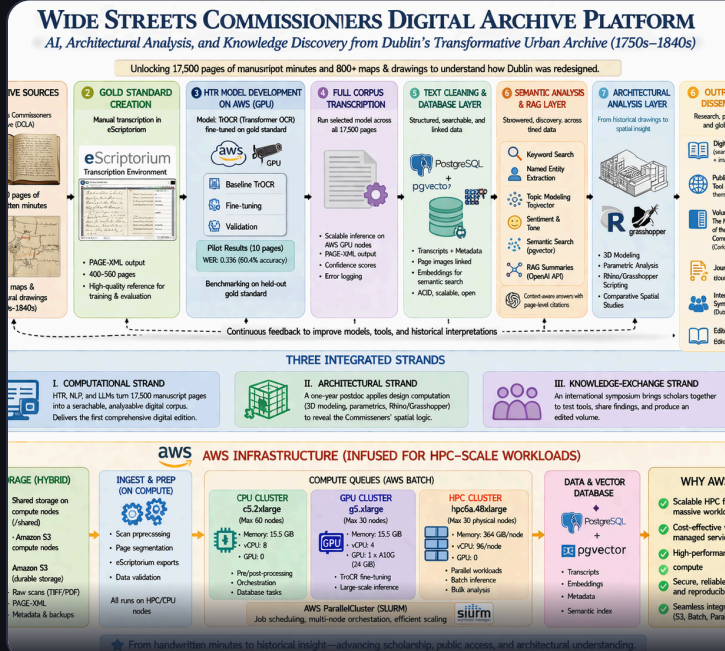
Django 5.2 PostgreSQL pgvector Sentence-Transformers Chart.js Semantic Search Render

► Technical deep-dive

[Live Demo](#) Login: demo / demo123
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AI & DATA ENGINEERING

Applied AI · Operations Research · Digital Humanities



APPLIED AI · DIGITAL HUMANITIES

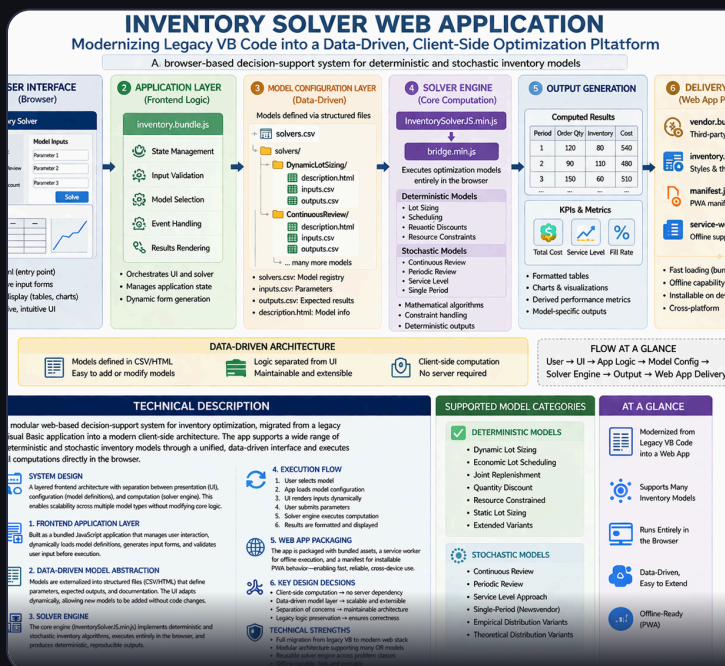
eScriptorium HTR Pipeline — Historical Manuscript Digitization

End-to-end Handwritten Text Recognition pipeline to digitize the Dublin Wide Streets Commissioners archive — 17,500 pages of 18th-century handwritten manuscripts. Achieved 86.4% character accuracy (CER 0.136) on pilot. Produced gold-standard PAGE-XML training datasets establishing the foundation for the first fully searchable digital edition of the archive.

eScriptorium HTR PAGE-XML BLLA segmentation Digital Humanities Python

► Pipeline overview

[Private Archive Project](#)



APPLIED OR · DECISION SUPPORT · WEB MIGRATION

Inventory Solver — Browser-Based Operations Research Platform

Full migration of a legacy Visual Basic inventory management system into a modern, client-side decision-support web application. Supports 10+ deterministic and stochastic OR models — including dynamic lot sizing, joint replenishment, quantity discount, continuous/periodic review, and newsvendor — with all computation executing entirely in the browser. Deployed as an offline-capable PWA with zero backend dependency.

JavaScript Operations Research PWA Service Workers OR Solver Engine Data-Driven Config

► Architecture deep-dive

[Launch App](#)

ABOUT

Ahmad Farhat

Data engineer, machine learning practitioner, and AI systems architect with expertise spanning the full ML lifecycle — from distributed data pipeline design and large-scale processing to model training, productization, and MLOps. Experienced deploying production AI platforms, institutional analytics systems, and research-grade ML infrastructure across cloud and on-premise environments.

Active researcher with peer-reviewed publications in Scopus-indexed Q1 journals — [Google Scholar ↗](#)